COURSE 3 · PHYSICS

Physics

Visual companion and Symbols Sheets

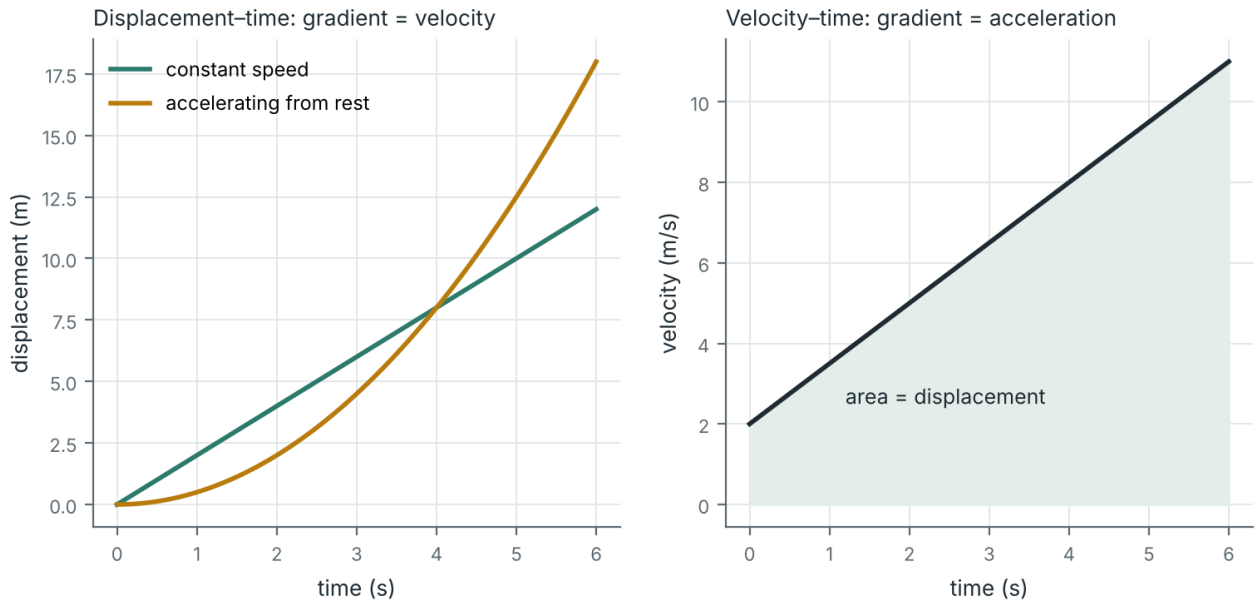
One section per module: the Symbols Sheet, with every numbered equation from the audio reader in ordinary notation, then the module's key diagram.

Keep it open while you listen, and beside you for the workbook.

Module 3.1 · Symbols and diagrams

Motion (kinematics)

3.1-1	Constant-acceleration equations	$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \frac{(u+v)t}{2}$
3.1-2	Launch components	$u_x = u\cos\theta, \quad u_y = u\sin\theta$
3.1-3	Range on level ground	$R = \frac{u^2 \sin 2\theta}{g}$

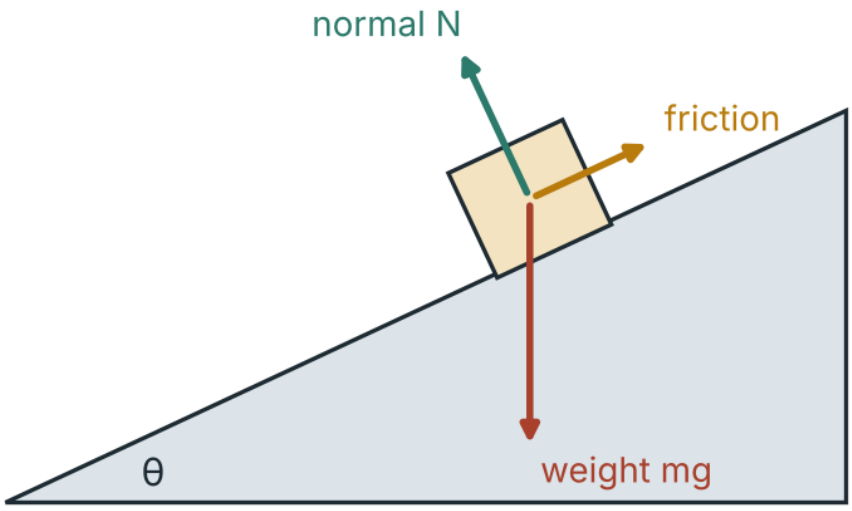


Motion graphs. The gradient of displacement–time is velocity; the area under velocity–time is displacement.

Module 3.2 · Symbols and diagrams

Forces and Newton's laws

3.2-1	Newton's second law	$F_{\text{net}} = ma$
3.2-2	Weight	$W = mg$
3.2-3	Friction	$F_f \leq \mu N$
3.2-4	Centripetal acceleration and force	$a = \frac{v^2}{r}, \qquad F = \frac{mv^2}{r}$



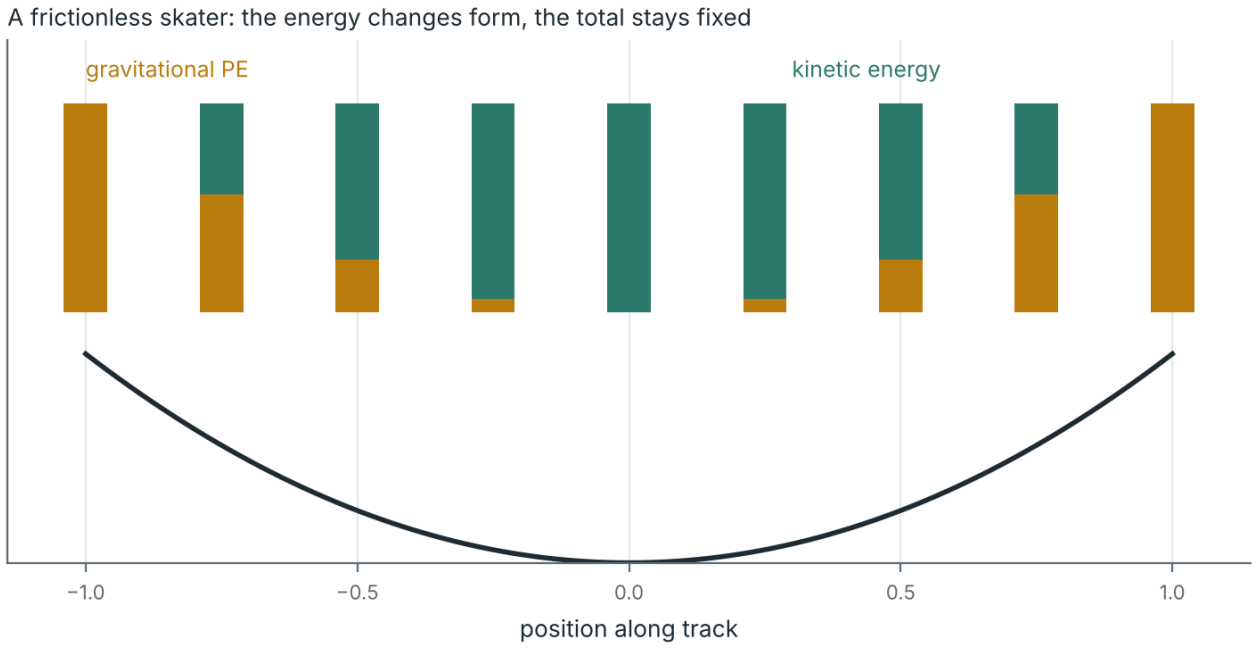
Weight splits into mg sin θ down the slope and mg cos θ into it.

Free-body diagram for a block on a slope: weight, normal force and friction.

Module 3.3 · Symbols and diagrams

Energy, work and power

3.3-1	Work	$W = Fd\cos\theta$
3.3-2	Kinetic energy	$E_k = \frac{1}{2}mv^2$
3.3-3	Gravitational potential energy	$E_p = mgh$
3.3-4	Power	$P = \frac{E}{t} = Fv$
3.3-5	Efficiency	$\eta = \frac{E_{\text{useful}}}{E_{\text{in}}} \times 100\%$



Energy bars for a frictionless skater: potential energy (ochre) converts to kinetic (teal) and back, total unchanged.

Module 3.4 · Symbols and diagrams

Momentum, collisions and rotation

3.4-1	Momentum	$p = mv$
3.4-2	Impulse	$F\Delta t = \Delta p$
3.4-3	Conservation of momentum	$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$
3.4-4	Torque	$\tau = Fr_\perp$
3.4-5	Angular momentum	$L = I\omega$

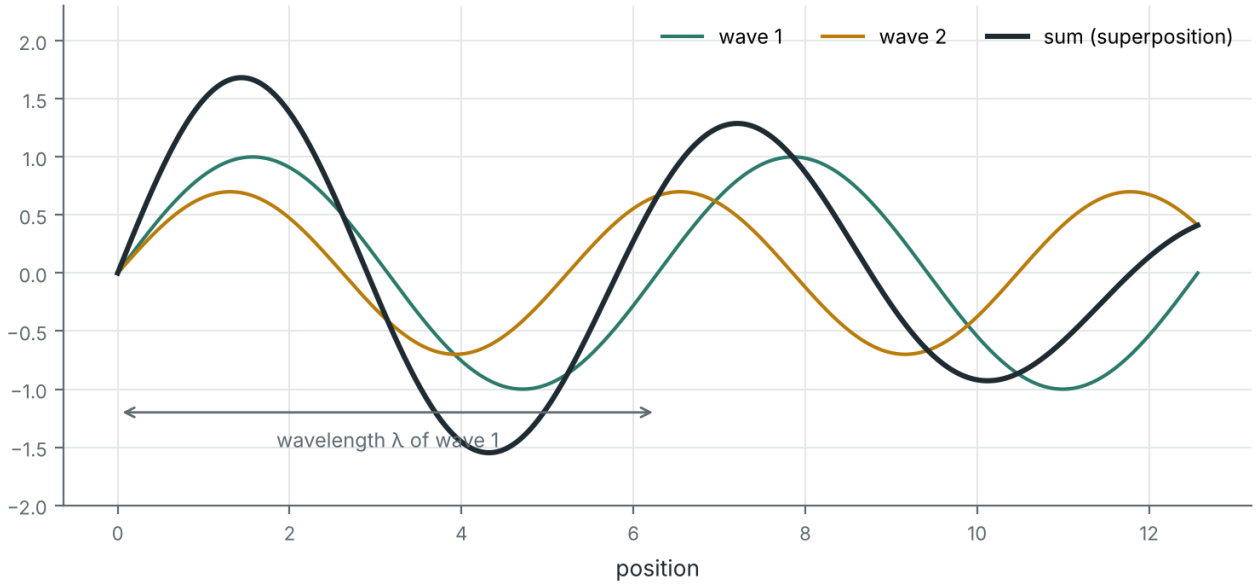


A perfectly inelastic collision: momentum before equals momentum after.

Module 3.5 · Symbols and diagrams

Waves and sound

3.5-1	Wave equation	$v = f\lambda, \qquad T = \frac{1}{f}$
3.5-2	String fixed at both ends	$\lambda_n = \frac{2L}{n}, \qquad f_n = \frac{nv}{2L}$
3.5-3	Doppler, moving source	$f' = f\frac{v}{v \mp v_s}$

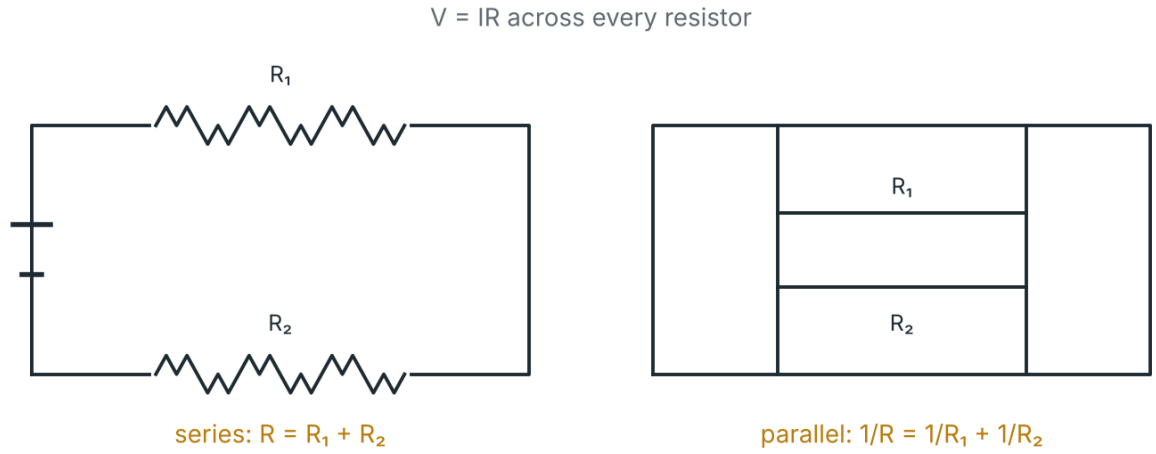


Superposition: two waves add point by point.

Module 3.6 · Symbols and diagrams

Electricity and circuits

3.6-1	Current	$I = \frac{Q}{t}$
3.6-2	Potential difference	$V = \frac{E}{Q}$
3.6-3	Ohm's law	$V = IR$
3.6-4	Series resistance	$R = R_1 + R_2 + \cdots$
3.6-5	Parallel resistance	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$
3.6-6	Electrical power	$P = VI = I^2R = \frac{V^2}{R}$

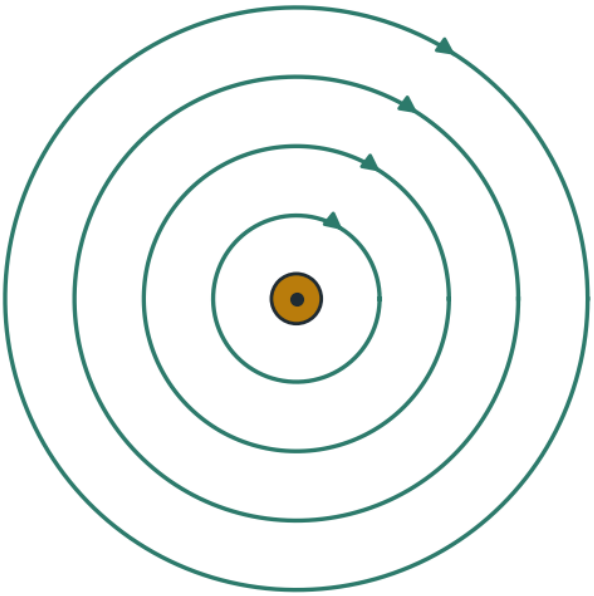


Series and parallel resistors.

Module 3.7 · Symbols and diagrams

Magnetism and electromagnetism

3.7-1	Force on a current	$F = BIL\sin\theta$
3.7-2	Force on a moving charge	$F = qvB\sin\theta$
3.7-3	Faraday's law	$\varepsilon = -N\frac{\Delta\Phi}{\Delta t}$
3.7-4	Ideal transformer	$\frac{V_s}{V_p} = \frac{N_s}{N_p}$



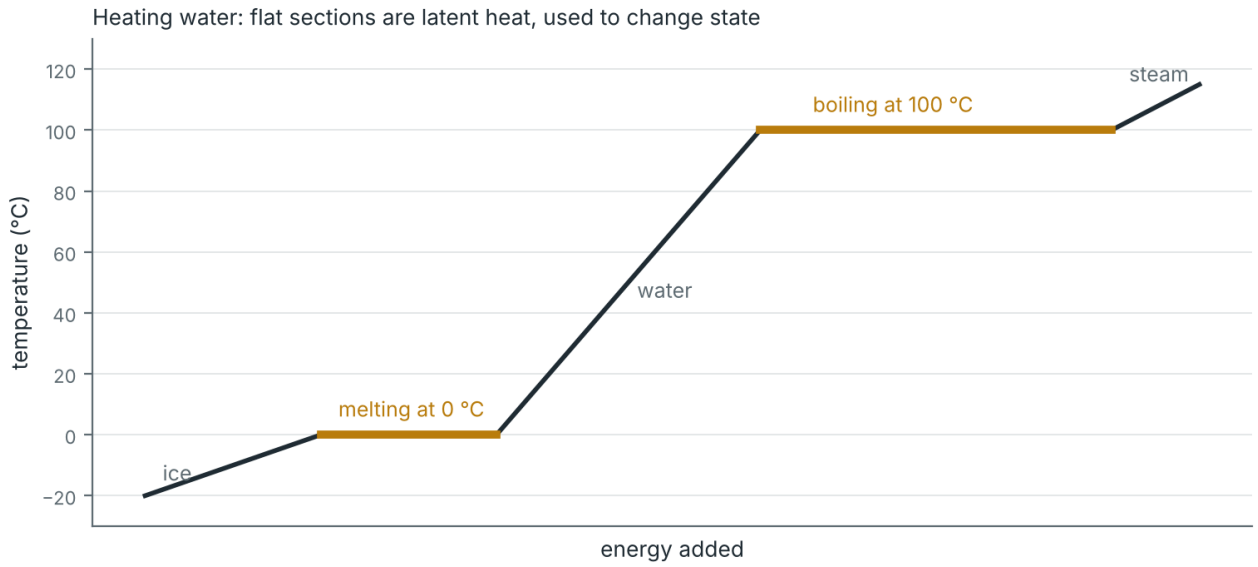
Current out of the page: field lines circle the wire (right-hand grip rule)

The magnetic field around a straight current-carrying wire forms concentric circles.

Module 3.8 · Symbols and diagrams

Thermodynamics and heat

3.8-1	Kelvin and Celsius	$T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273$
3.8-2	Specific heat	$Q = mc\,\Delta T$
3.8-3	Latent heat	$Q = mL$
3.8-4	First law	$\Delta U = Q - W$
3.8-5	Maximum (Carnot) efficiency	$\eta_{\max} = 1 - \frac{T_c}{T_h}$

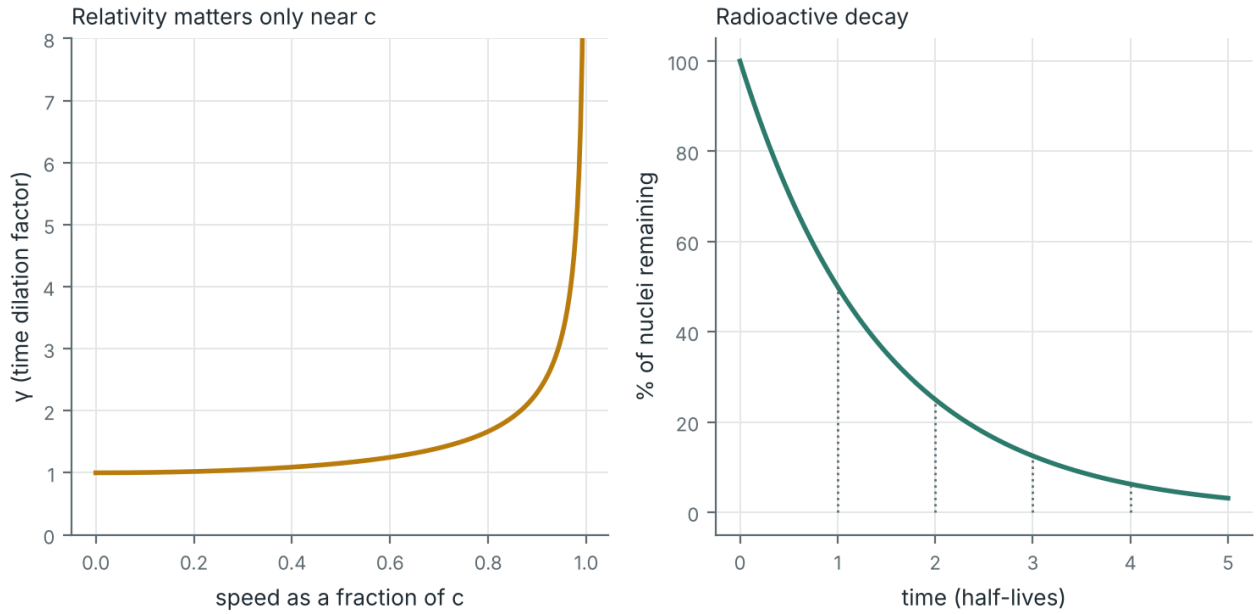


Heating curve of water: temperature plateaus while the state changes.

Module 3.9 · Symbols and diagrams

Modern physics

3.9-1	Lorentz factor and time dilation	$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}, \qquad t = \gamma t_0$
3.9-2	Mass–energy equivalence	$E = mc^2$
3.9-3	Photon energy	$E = hf = \frac{hc}{\lambda}$
3.9-4	Spectral line	$\Delta E = E_{\text{upper}} - E_{\text{lower}} = hf$
3.9-5	Radioactive decay	$N = N_0 \left(\frac{1}{2}\right)^{t/t_{1/2}}$



Left: the time dilation factor stays near 1 until speeds approach light speed. Right: radioactive decay halves the remaining nuclei every half-life.

Constant	Symbol	Value
speed of light	c	3.00×10^{8} m/s
Planck constant	h	6.63×10^{-34} J s
electron charge	e	1.60×10^{-19} C
electron volt	eV	1.60×10^{-19} J